

AI-Based Crop Disease Detection System Using Deep Learning

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Project Report

Technology Used: Python, TensorFlow, CNN, Streamlit, Google Colab

Introduction

Crop diseases significantly affect agricultural productivity and food quality. This project uses Deep Learning and Convolutional Neural Networks (CNN) to detect crop diseases from leaf images automatically.

Objectives

- Detect crop diseases using AI
- Provide faster and accurate disease prediction
- Help farmers with treatment suggestions
- Reduce manual inspection effort

Technologies Used

- Python
- TensorFlow / Keras
- Streamlit
- Google Colab
- CNN Deep Learning Model

Dataset

The project uses the PlantVillage dataset containing multiple classes of healthy and diseased crop leaves. The images are preprocessed and used for training and validation.

Model Architecture

The CNN model consists of convolution layers, max pooling layers, flatten layers, dense layers, and dropout layers for improved accuracy and reduced overfitting.

Website Interface

A Streamlit-based web application was developed where users can upload crop leaf images and receive disease prediction, confidence score, and treatment suggestions.

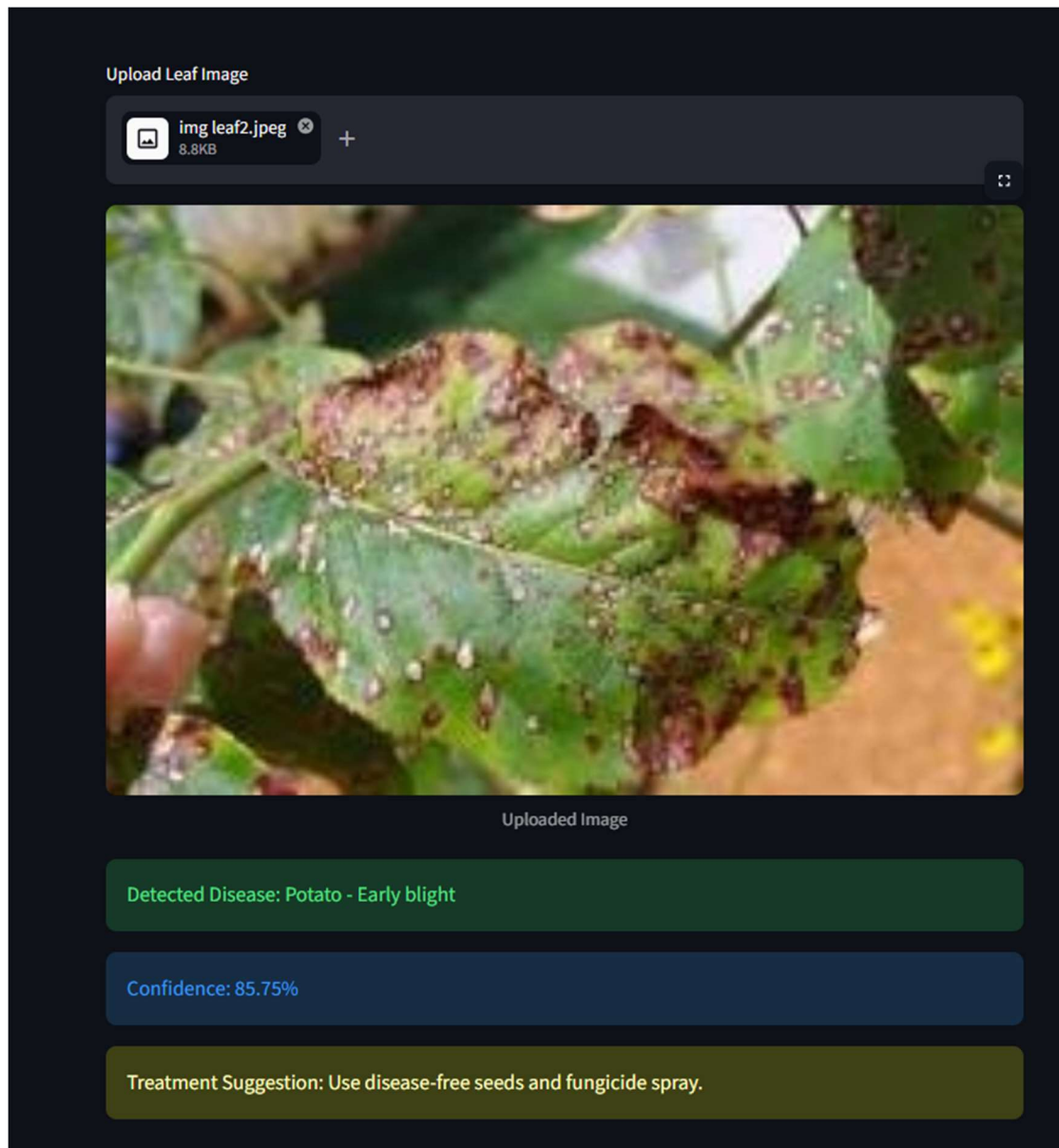
Results

The trained model achieved high accuracy during validation and successfully predicted multiple crop diseases from uploaded leaf images.

Conclusion

The AI-Based Crop Disease Detection System provides an efficient and accurate solution for identifying plant diseases and supporting smart agriculture practices.

Project Screenshots



Crop Disease Detection Website Output 1

Upload a leaf image to detect crop disease.

Upload Leaf Image



img2.jpeg
9.0KB



Uploaded Image

Detected Disease: Potato - Early blight

Confidence: 98.74%

Treatment Suggestion: Use disease-free seeds and fungicide spray.

Crop Disease Detection Website Output 2

Model: "sequential"

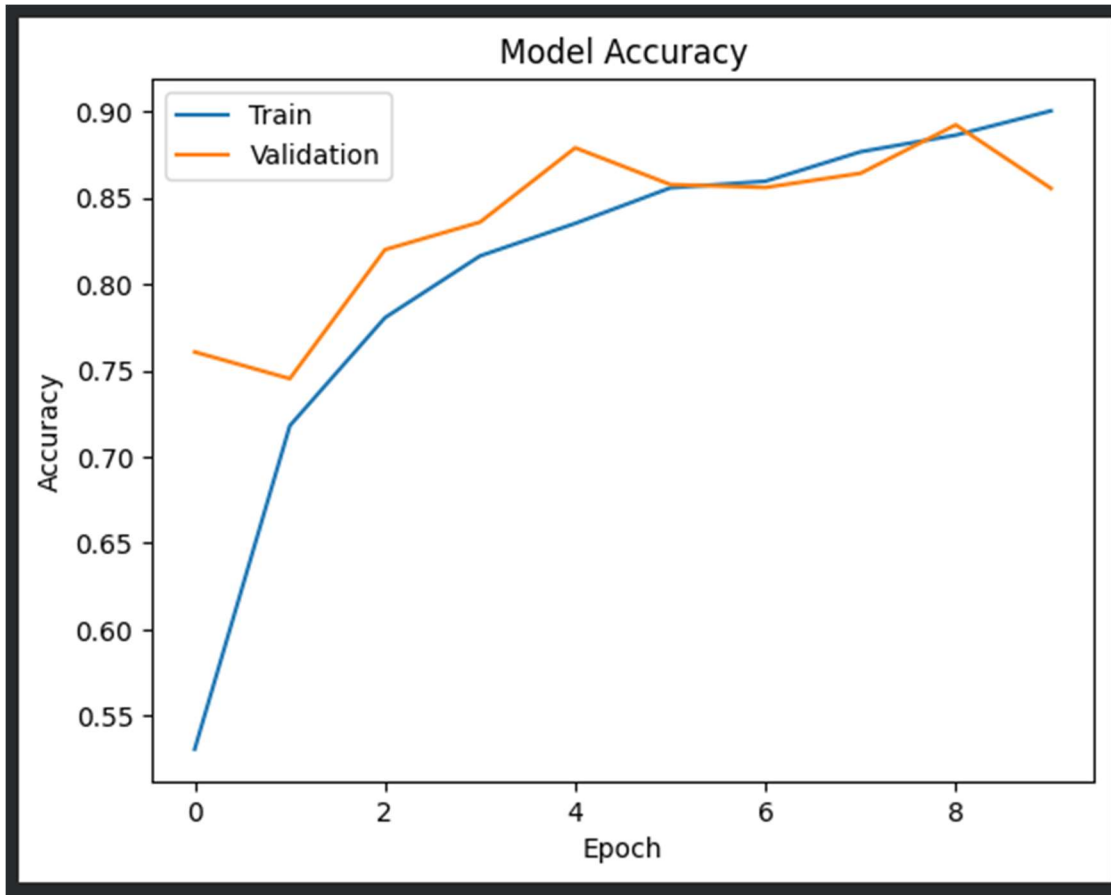
Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 222, 222, 32)	896
max_pooling2d (MaxPooling2D)	(None, 111, 111, 32)	0
conv2d_1 (Conv2D)	(None, 109, 109, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 54, 54, 64)	0
conv2d_2 (Conv2D)	(None, 52, 52, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 26, 26, 128)	0
flatten (Flatten)	(None, 86528)	0
dense (Dense)	(None, 128)	11,075,712
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 15)	1,935

Total params: 11,170,895 (42.61 MB)

Trainable params: 11,170,895 (42.61 MB)

Non-trainable params: 0 (0.00 B)

CNN Model Architecture



Model Accuracy Graph

Future Scope

- Mobile Application Integration
- Real-time Camera Detection
- Multi-language Support
- Cloud Deployment
- Weather-based Disease Analysis

References

- TensorFlow Documentation
- Streamlit Documentation
- PlantVillage Dataset
- Deep Learning Research Papers